

$$\begin{array}{rcll} & 100 & 90 & \\ & 5 & 5 & 25 \\ \text{[points=5]} & & & 1, 3, 6, 10, 15, 21, 28, 36, \dots & 50 \quad \text{[A]} \\ 1275 & & & & \\ 1596 & & & & \\ 1597 & & & & \\ 1598 & 2, 3, 4, \dots & & & \end{array}$$

$$a_{50} = 50 \times 512 = 1275 \quad \text{A}$$

$$\text{[points=5]} \quad \{a_n\}$$

$$\begin{array}{l} a_n < 930 \quad n \quad \text{[B]} \\ 28 \\ 29 \\ 30 \\ 31 \end{array}$$

$$29 \times 30 = 870 < 930 \quad 30 \times 31 = 930 \quad 29 \quad \text{B}$$

$$\text{[points=5]} \quad \{a_n\} \quad a_1 = 4$$

$$\begin{array}{l} a_n = \text{[D]} \\ n \\ 2n \\ 3n \\ 4n \quad b_n = a_n - 4n \end{array}$$

$$\begin{array}{l} b_1 = a_1 - 4 = 0 \quad b_n = 0 \quad a_n = 4n \quad \text{D} \\ \text{[points=5]} \quad \{a_n\} \quad a_1 = 1 \quad a_n + a_{n+1} = 2n \quad \{a_n\} \quad \text{[B]} \\ a_n = n \\ a_n = \{n, n, \\ n-1, n \\ a_n = \{n, n, \\ n+1, n \\ a_n = \{n-1, n, \\ n+1, n \end{array}$$

$$a_{n+2} - a_n = 2 \quad a_1 = 1 \quad a_2 = 1$$

$$\begin{array}{l} \text{B} \\ \text{[points=5]} \quad \{a_n\} \quad n \end{array}$$

$$\begin{array}{l} b_n = 1a_n a_{n+1} \quad T_n \quad \{b_n\} \quad n \quad T_n = \text{[B]} \\ 13 - 13n + 1 \\ 13 - 13(3n + 1) \\ 13 - 13(3n - 1) \\ 13 + 13(3n + 1) \quad a_n = S_n - S_{n-1} \quad n = 1 \quad a_n = 3n - 2 \end{array}$$

$$\begin{array}{l} \text{B} \\ 2 \quad 6 \quad 12 \quad 6 \quad 0 \\ \text{[points=6]} \quad \{a_n\} \quad a_1 = -2 \end{array}$$

$$\begin{array}{l} n \quad S_n \quad \text{[ABD]} \\ a_2 = -8 \\ a_n = -n \cdot 2^n \\ S_3 = -30 \\ S_n = (1 - n)2^{n+1} - 2 \end{array}$$